



Society of Petroleum Engineers

SPE-227751-MS

Evaluating the Effect of Capillary Pressure on Rock Mechanical Properties

Shujath A. Syed, Baker Hughes, Khobar, Saudi Arabia; S. Mark Ma, Saudi Aramco, Dhahran, Saudi Arabia; S. Al-Ofi and J. Franquet, Baker Hughes, Khobar, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227751-MS](https://doi.org/10.2118/227751-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Geo-mechanical models targeting downhole applications always consider effective stress in terms of pressure of pore fluid, typically single phase; dry or wet. To simulate reservoirs in situ where multiphase fluids co-exist, other effects such as capillary pressure, need to be taken into consideration. The objective of this study is to evaluate the effect of capillary pressure on the effective stress in a laboratory environment.

Effective stress is mainly considered a function of pore fluid pressure and Biot's coefficient. To measure Biot's coefficient and effective stress, hydrostatic compression tests were performed on sandstone rocks saturated with a single phase fluid; brine or oil. To study effect of capillary pressure of multiphase systems on effective stress, a second fluid, air or oil, was introduced into the water saturated rock by centrifugation. When reaching irreducible water saturation, then capillary pressure was recorded. Effective stress was again measured on the multiphase samples and compared with that of single-phase systems, to quantify the effect of capillary pressure on effective stress.

In multiphase systems (particularly air/brine), the Biot coefficient displayed complex stress-dependent evolution, initially decreasing to a minimum before increasing again. This nonlinear trend likely reflects pore compression followed by collapse. In contrast, oil/brine systems primarily showed a decreasing Biot coefficient with stress, only rising slightly at the end of the loading cycle. Compared to single-phase brine, multiphase systems (air/brine and oil/brine) exhibited lower Biot coefficients, suggesting capillary pressure enhances effective stress by compressing the rock matrix. These findings are critical for reservoir characterization, Geo-mechanical modeling, and production forecasting, particularly in gas reservoirs. Notably, the Biot coefficient's nonlinear response to applied loading—observed in Nugget and Berea sandstones—highlights its dependence on effective compression and porosity reduction.

The novelty of the present workflow lies in challenging the current approach of estimating effective stress on rocks which traditionally assumes single phase fluid, with experimental results of evaluating and quantifying effect of capillary pressure on effective stress measurements and Biot coefficient. Future work should explore a wider range of fluid systems and integrate in-situ imaging to elucidate pore-scale mechanisms.

Introduction

Understanding rock mechanical behaviors is fundamental to a wide range of Geo-mechanical applications, including wellbore stability, hydraulic fracturing, and sand production (Syed et al., 2022; Ali et al., 2024). Traditionally, the effective stress concept introduced by Terzaghi (1936 & 1943) and Biot (1941) has served as the cornerstone for modeling rock mechanical properties in these contexts. Terzaghi's effective stress (σ_e) theory postulates that the mechanical behavior of a porous rock is governed by the difference between the total stress (σ) and the pore fluid pressure (p_f), effectively treating the pore pressure as a uniform, isotropic force acting against the confining stress. This approach, while powerful, assumes that the pore pressure is evenly distributed and does not account for the complex interactions at the pore scale, particularly those arising from capillary pressure (P_c) forces existing in formations where multiphase fluids coexist from oil migration to waterflooding oil recovery (Al-Harbi et al., 2011).

Capillary pressure P_c is defined as the pressure proportional to the curvature of an interface between two immiscible fluids, or inversely proportional to the principal radii of the curvature (Ma et al., 1991). It is conventionally expressed as the pressure difference between nonwetting phase such as oil (P_o) and wetting phase water (P_w) pressure in an oil reservoir, i.e., $P_c = P_o - P_w$, and it becomes especially significant in rocks with structure of small pores, such as tight formations or shales, where, P_c can reach magnitudes that are comparable to the system stresses, fundamentally altering the distribution and behavior of fluids within the rock matrix. This, in turn, can have a pronounced impact on the mechanical properties of the rock, including its stiffness, strength, and failure characteristics (Bui & Tutuncu, 2015; Moghadam et al., 2020; Albannay et al., 2021).

Recent experimental and modeling studies have demonstrated that P_c can reinforce grain contacts through capillary condensation, leading to measurable increases in Young's modulus and changes in acoustic properties as fluids condense within nanopores (Bui & Tutuncu, 2015; Albannay et al., 2021). For example, capillary condensation in shale samples has been shown to increase Young's modulus by up to 10%, with the effect becoming more pronounced at higher stress levels. These changes are not captured by the conventional effective stress models (Terzaghi, 1936 & 1943; Biot, 1941), which ignore the role of P_c and thus may underestimate the true mechanical response of the rock under reservoir conditions (Bui & Tutuncu, 2015; Moghadam et al., 2020).

The implications of neglecting P_c in Geo-mechanical models can be far-reaching. In wellbore stability analysis, for instance, the omission of P_c effects can lead to inaccurate predictions of rock strength and failure, potentially resulting in wellbore collapse or excessive deformation. Similarly, in hydraulic fracturing, P_c influences the closure and conductivity of fractures, affecting both the initiation and propagation of fractures as well as the efficiency of proppant placement (Moghadam et al., 2020; Zhang & Yang, 2024). During sand production, P_c can alter the mobilization threshold of grains, impacting the onset and severity of sanding events.

Moreover, advanced coupled flow-geomechanics models have begun to incorporate P_c effects, revealing that dynamic changes in fracture aperture and P_c can significantly influence fluid flow, fracture deformation, and production rates in unconventional reservoirs (Haghi & Chalaturnyk, 2022; Yoon & Kim, 2023; Zhang & Yang, 2024). These models highlight the necessity of accounting for P_c not only in laboratory measurements of rock mechanical properties but also in field-scale Geo-mechanical studies. The main objective of the present study is to try to quantify the contribution of P_c towards effective stress measurements by performing laboratory experimentation.

Effective Stress Theory- Rock/Fluid Systems

Effective stress theory is central to understanding the mechanical behavior of poroelastic rocks. The theory, originally formulated by Terzaghi (1936 & 1943) and later extended by Biot (1941), postulates that the deformation and strength of a porous rock are governed not by the total applied stress (σ), but by the effective

stress (σ_e), defined as the difference between the external applied stress and a fraction of the pore fluid pressure (p_f), a single phase fluid, as shown in Equation 1.

$$\sigma_e = \sigma - \alpha_s p_f \quad (1)$$

This fraction is quantified as the Biot coefficient (α) or more specifically α_s where the subscript s is for single phase fluid. Hereon, α or α_s is used interchangeably to indicate Biot coefficient which is usually measured using single phase fluid saturation.

$$\alpha_s = \frac{\sigma - \sigma_e}{p_f} = 1 - \frac{K_r}{K_m} \quad (2)$$

where K_r is the bulk modulus of the rock (or frame) and K_m is the bulk modulus of the solid mineral phase.

The Biot coefficient α is a fundamental poroelastic parameter that quantifies the coupling between applied stress and pore pressure in a porous medium. Several experimental and theoretical approaches can be employed to determine α (Detournay & Cheng, 1993; Hart & Wang, 1995; Mavko et al., 2009), which reflects the degree to which pore pressure counteracts the applied load on the rock skeleton (Kümpel, 1991; Guerriero & Mazzoli, 2021; Kasani & Selvadurai, 2022). Physically this means that the solid frame of rock matrix carries the σ_e part of σ while the remaining part (αp_f) is carried by the fluid.

$$\sigma = \sigma_e + \alpha_s p_f \quad (3)$$

The rest of the pore pressure, $(1-\alpha)p_f$, is counteracted by the internal stresses of the rock. Note that at its minimum limit of $\alpha=0$, p_f does not contribute to σ at all, such as in an open system, while at its maximum limit of $\alpha=1$, p_f does its best to counter total stress σ . It should also be noted that the total stress σ acting of the rock matrix is transferred through the contact points of the individual particles, thus rock fails when the effective stress σ_e , the stress acting on the solid matrix particles, exceeds the strength of solid matrix.

Capillary Pressure Effect- Rock/Multiphase Fluid Systems

In multiphase fluid systems, such as hydrocarbon reservoirs, capillary pressure P_c plays a significant role in the mechanical and flow behavior of partially saturated rocks. To accurately describe these effects, P_c must be incorporated into the effective stress theory when dealing with multiphase problems, as described by Bishop (1959), Papamichos et al. (1997), and Maranha et al. (2022).

As it is known that the interfacial tension between wetting and non-wetting phase, the pore structure, and wettability create a lower pressure in the wetting phase (P_w) surrounding grains than the pore fluid pressure (p_f) or oil pressure (P_o), as illustrated in Figure 1. The effective stress can then be modified as (Arii et al, 2005):

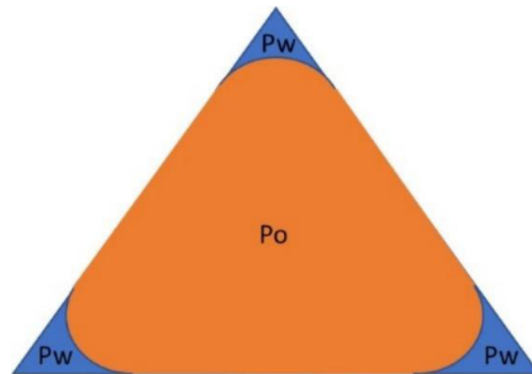


Figure 1—Schematic of oil and water distribution in a triangular pore (Ma et al., 1996 and 2006)

$$\sigma_e = \sigma - \alpha_m (P_f - P_c) \quad (4)$$

Where, α_m is Biot coefficient measured for multiphase fluid saturation and capillary pressure P_c is defined as:

$$P_c = P_o - P_w = p_f - P_w \quad (5)$$

Eqs 1 and 4 form the base to study the effect of capillary pressure on rock mechanical properties such as the Biot coefficient.

Experimental Methodology

The experimental approach in this study is mainly based on the effective stress measurements, wherein it is expressed as a function of total stress (σ), pore pressure (P_f), Biot's coefficient (α) for the case of single-phase fluid (Eq. 1), and also capillary pressure (P_c) for the two phase system, as given in Eq. 4. This formulation allows for the explicit inclusion of P_c effects, which is essential for rocks containing multiple fluid phases, such as gas and oil reservoirs with connate water saturation. To measure effective stress, Matching Volumetric Strain (MVS) method is employed to determine the Biot coefficient, while P_c effects are assessed by comparing single-phase (fully saturated) and two-phase (air/brine or oil/brine partially saturated) fluid systems. Three types of sandstone with varying porosities and permeabilities are used to generalize findings across different rock qualities.

Sample Preparation and Characterization

Three outcrop sandstones were selected to cover different range of porosity (ϕ) and permeability (k) as shown in Table 1. This diversity ensures that the results capture the variability in mechanical and petrophysical behaviors associated with different rock textures and pore structures. Cylindrical cores of 1.5" (38 mm) in diameter and 3" (76 mm) in length were drilled and polished to minimize end effects. To preserve clay structures, samples were oven-dried at 60°C for 6 hours to remove residual moisture. Porosity was measured using helium porosimeter, while permeability was determined via steady-state flow tests. Also, using weight method, porosity of samples is calculated after full saturation with brine and oil and compared to measured helium porosity as shown in Table 1.

Table 1—Porosity and permeability of selected samples

Sandstone Name	Permeability (mD)	Porosity (%)		
		Helium	Weight from Brine Saturation	Weight from Oil Saturation
Nugget	2.9	17.8	14.9	14.9
Berea	190	19.29	18.3	18.1
Torrey Buff	298	13.93	13.2	13.2

From Table 1, it is noted that porosities measured using the two methods are similar for the two samples with high permeabilities, though the helium porosity is slightly higher by about 1 pu. The difference between the two methods is more pronounced for the low permeability Nugget sample which may attribute to the difficulties in completely saturating low permeability samples with the procedures adopted.

Fluid Saturation Protocols

Initially, all samples were fully saturated with a single-phase fluid (brine or oil) to establish baseline effective stress and Biot coefficient α_s measurements. This step provides a reference for subsequent tests involving multiphase fluids. Samples were fully saturated with 5% NaCl solution brine under vacuum, followed by a hydrostatic pressure of 2000 psi for 24 hours. For multiphase experiments, samples were first saturated with

brine and then subjected to controlled drainage using either air or oil as the non-wetting phase. The drainage process was performed in a centrifuge to achieve low, often called irreducible (Ma and Musharfi, 2020), water saturation, ensuring that the distribution of fluids within the pore space mimics reservoir conditions. Samples were centrifuged at 3000 rpm for air/brine system and at 4000 rpm for oil/brine system to displace brine with air or oil, respectively. Centrifugation was continued until no further change in water saturation was observed, confirming the attainment of "irreducible" saturation at the set centrifuge RPM for both air/brine and oil/brine systems. Centrifugation data with displaced volume and capillary pressure obtained is shown in Table 2.

Table 2—Centrifugation data for the 3 cores for both air/brine and oil/brine systems

Sandstone Plug Name	He Porosity	Permeability	RPM	Calc Pc	Swir (g-w)		RPM	Calc Pc	Swir (o-w)
	%	mD		psi (MPa)	%PV			Psi (MPa)	%PV
Nugget	17.8	2.9	3000	207 (1.5)	40.7		4000	100 (0.7)	53.2
Berea	19.29	190	3000	207 (1.5)	15.9		4000	100 (0.7)	42.3
Torrey Buff	13.93	298	3000	210 (1.5)	18.5		4000	101 (0.7)	43.3

Matching Volumetric Strain (MVS) Method for Biot Coefficient Determination

The MVS method is an experimental technique used to determine α by comparing volumetric strains under different loading conditions of drained and undrained. Unlike direct measurements of K_m (Eq. 2), which are challenging. The MVS method provides an indirect, and yet still accurate, estimation of α by matching strain responses (Franquet & Abass, 1999), relying on two key experiments:

- Drained (D) hydrostatic compression test which directly measures effective stress σ_e^D in absence of pore pressure ($p_f = 0$ in Eq. 1).
- Undrained (UD) hydrostatic compression test which measures total stress σ^{UD} at a constant applied pore pressure (p_f^{UD}).

By comparing the volumetric strains (ε_v) from these tests, α_s can be derived using Eq. 2, without requiring direct measurement of K_s .

$$\alpha_s = \frac{\sigma^{UD} - \sigma_e^D}{p_f^{UD}} \quad (6)$$

Figure 2 shows the confining and pore pressure cycles applied to measure Biot's coefficient. During drained compression test (cycles 1 & 2), a jacketed sample fully saturated with liquid is subjected to incremental confining pressure, i.e., hydrostatic stress ($\sigma_1 = \sigma_2 = \sigma_3$), while $p_f = 0$ as the sample is allowed to drain. Thus, the applied confining pressure acts as effective stress (Eq. 1). Volumetric strain (ε_v) is recorded using linear variable differential transformers (LVDTs). As both the drained and undrained hydrostatic compression tests are performed on the same sample, loading and unloading processes can have significant effects on the rock properties (Morrow et al., 1990) including ε_v . Unloading can result in irreversible plastic volumetric strain, especially after the sample has undergone significant compaction during loading process. Therefore, for the first drained compression test, two cycles of loading are performed, with the aim of first cycle to evaluate the hysteresis effect as shown in Figure 2. Data from second cycle (cycle2) is used for interpretation of the drained compression test.

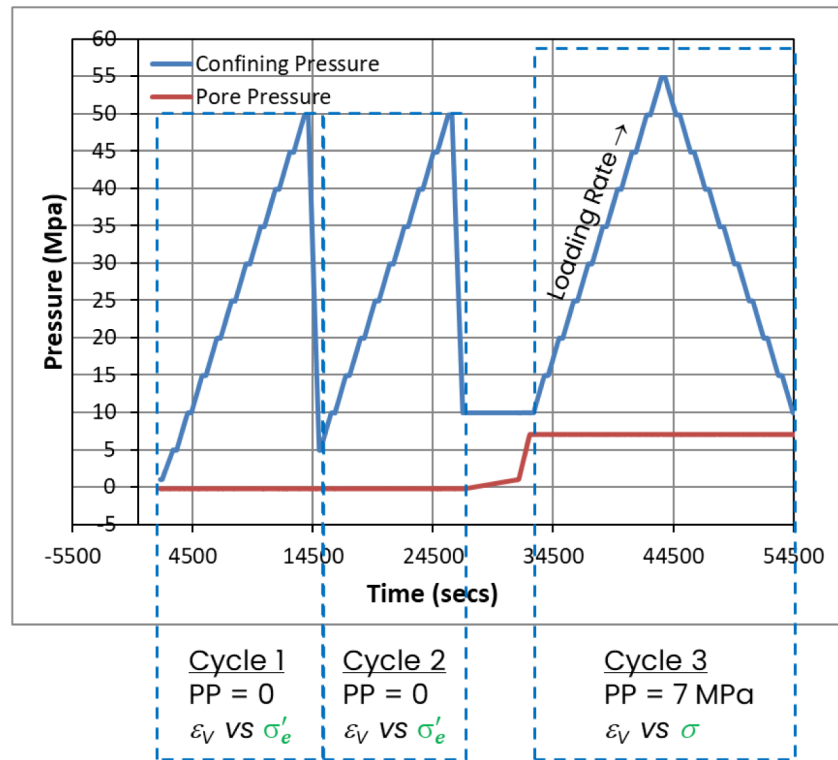


Figure 2—Test procedure for Biot Coefficient measurement

During undrained compression test, the sample is fully saturated with a liquid (brine or oil) and is subjected to an incremental confining pressure while the pore pressure (p_f or PP) is maintained at a constant value (7 MPa in this study) as shown for cycle3 in Figure 2. The applied confining pressure acts as a total stress σ . At the same ε_v , during both the drained and undrained tests, Biot's coefficient can be calculated using Eq. 6 as shown in Figure 3.

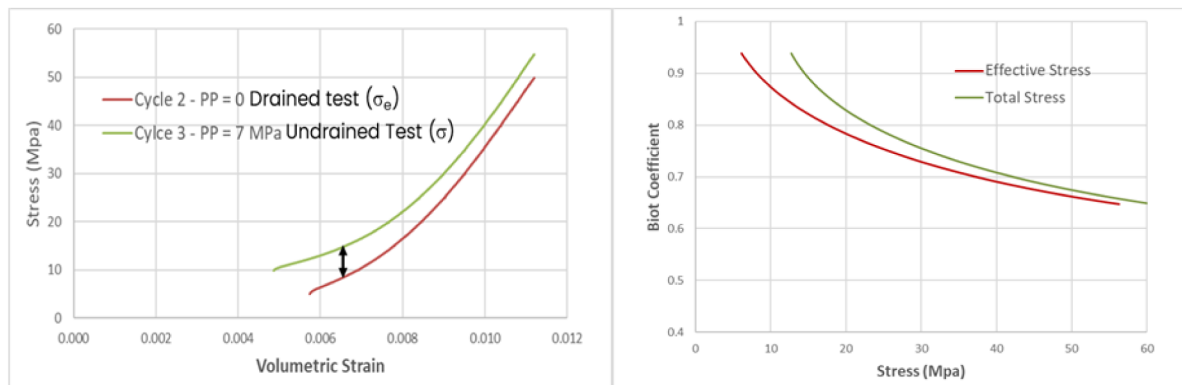


Figure 3—Sample data for MVS method to measure effective stress and Biot's coefficient.

Below summarizes the MVS methodology.

1. Select samples of sandstone outcrops with varying rock properties of porosity and permeability.
2. Prepare cylindrical samples and measure initial properties.
3. Saturate samples with brine (single-phase).
4. Measure baseline effective stress (σ_e) and single-phase Biot coefficient (α_s) as using the MVS method.

5. Introduce air as non-wetting phase, centrifuge to irreducible water saturation, $S_{\text{wir-air}}$, and calculate $\max P_c$.
6. Perform effective stress measurements under two-phase (air/brine) conditions.
7. Soxhlet cleaning and saturate samples with brine (single-phase).
8. Introduce oil as non-wetting phase, centrifuge to irreducible water saturation, $S_{\text{wir-oil}}$, and calculate $\max P_c$.
9. Perform effective stress measurements under two-phase (oil and brine) conditions.
10. Analyze data to quantify effect of P_c on effective stress (σ_e) and multiphase Biot coefficient (α_m), Eq 4.

Results and Discussion

Benchmark Effective Stress Measurements with Single-Phase Tests

To establish a reference for subsequent experiments involving P_c effects, effective stress measurements were first performed on sandstone samples fully saturated with a single-phase fluid (brine). This baseline allowed for the isolation of mechanical responses attributable solely to the interplay between total stress (σ), pore pressure (p_f), and the intrinsic properties of the rock matrix, without the added complexity of multiphase fluid interactions. The MVS method, as detailed above, was employed for the determination of both volumetric strain and Biot coefficient across all samples. Figure 3 presents the relationship between volumetric strain and both total and effective stress for the tested sandstone samples. As anticipated, for a given volumetric strain, the total stress required is consistently higher than the corresponding effective stress. This observation is in line with the effective stress principle, according to which, the pore pressure component acts to reduce the net stress experienced by the rock skeleton (Terzaghi, 1943; Biot, 1941; Zimmerman, 1991). Within the range of testing conditions, volumetric strain increases monotonically with increasing applied stress, a trend that is consistent with the expected compaction behavior of porous sandstones under load. The difference between total and effective stress at each strain increment reflects the contribution of pore pressure, modulated by the Biot's coefficient, to the overall mechanical response of the rock.

Using the effective stress equation (Eqn. 2 or 6) and the measured values of total stress, effective stress and pore pressure, at same volumetric strain, the Biot's coefficient was calculated for each sample at various stress levels. Figure 4a, Figure 5a & Figure 6a illustrates the variation of the Biot's coefficient as a function of applied stress for the three sandstone samples. A clear trend emerges: the Biot coefficient decreases with increasing applied stress, again, within the test conditions. This behavior is well-documented in the literature and is attributed to the progressive closure of microcracks and pore throats and increase in grain contact under compressive load, which increases the stiffness of the rock matrix relative to the pore space (Zamani & Knez, 2023; Saurabh & Harpalani, 2018; Alam & Fabricius, 2013). As the rock becomes less compressible with increasing stress, a smaller fraction of the applied load is transferred to the pore fluid, resulting in a lower Biot's coefficient. Figure 7a shows the comparison of the single-phase Biot coefficient behavior with stress for the three selected samples. Biot coefficient was consistently lower for the least permeable Nugget sample, compared to Berea and Torrey Buff. For Nugget sample, Biot coefficient was ranging between 0.85 to 0.54 with increasing stress, while it was between 0.94 to 0.75 for Berea, similar to that of Torrey Buff ranging between 0.86 to 0.73.

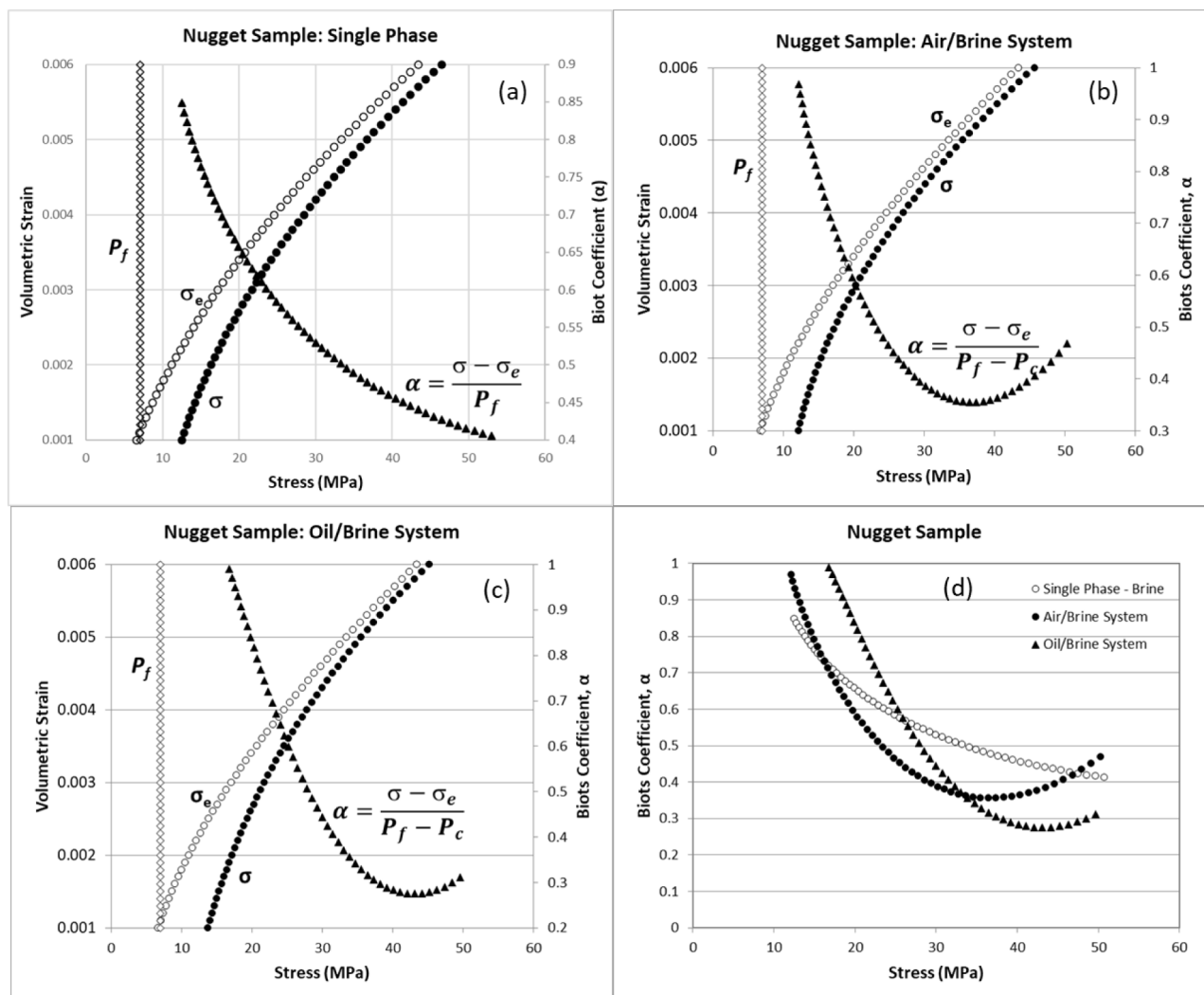


Figure 4—Measurement of effective stress, total stress and Biot coefficient for single phase brine, multi-phase air/brine and oil/brine system and their comparison for Nugget Sample.

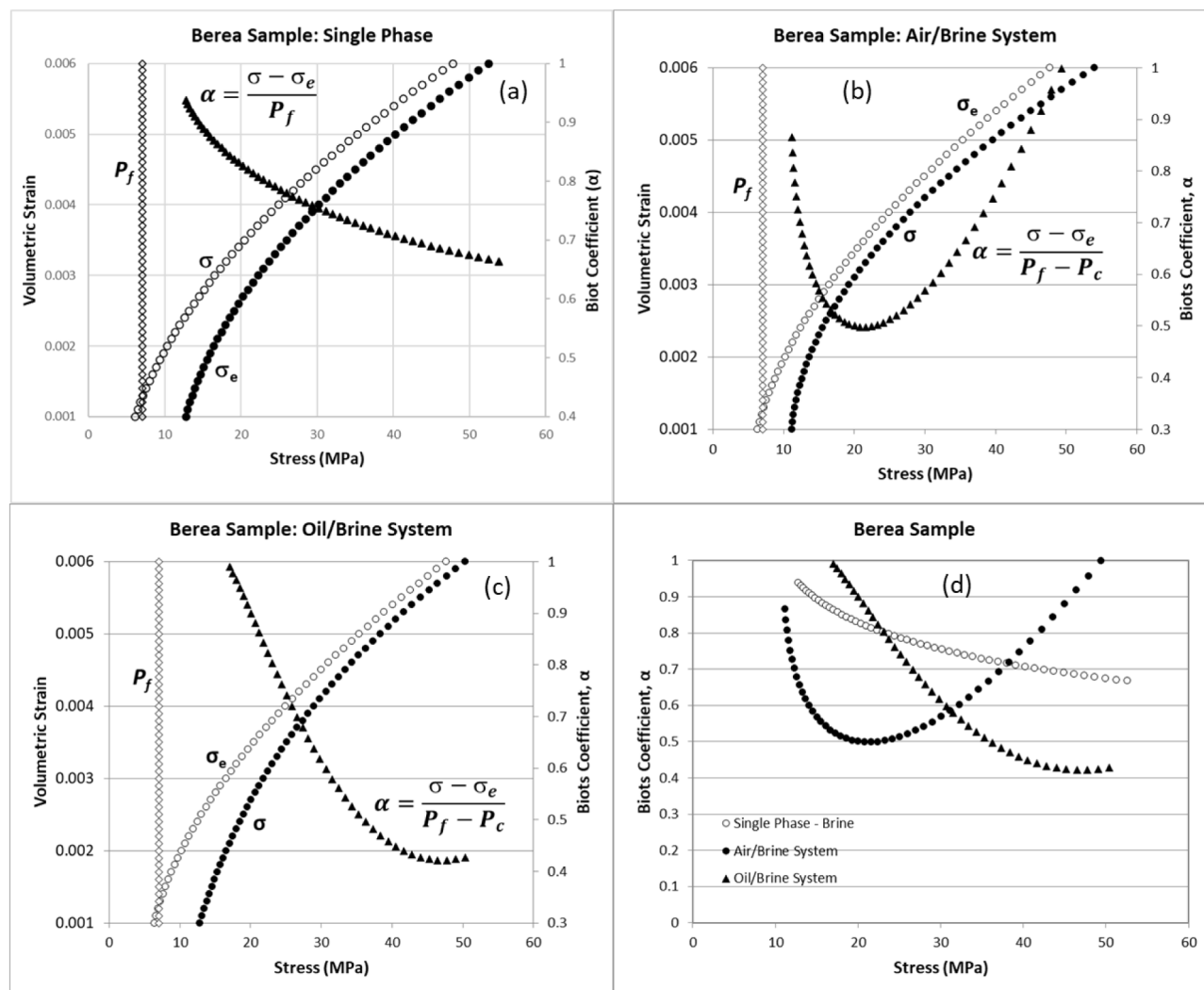


Figure 5—Measurement of effective stress, total stress and Biot coefficient for single phase brine, multi-phase air/brine and oil/brine system and their comparison for Berea Sample.

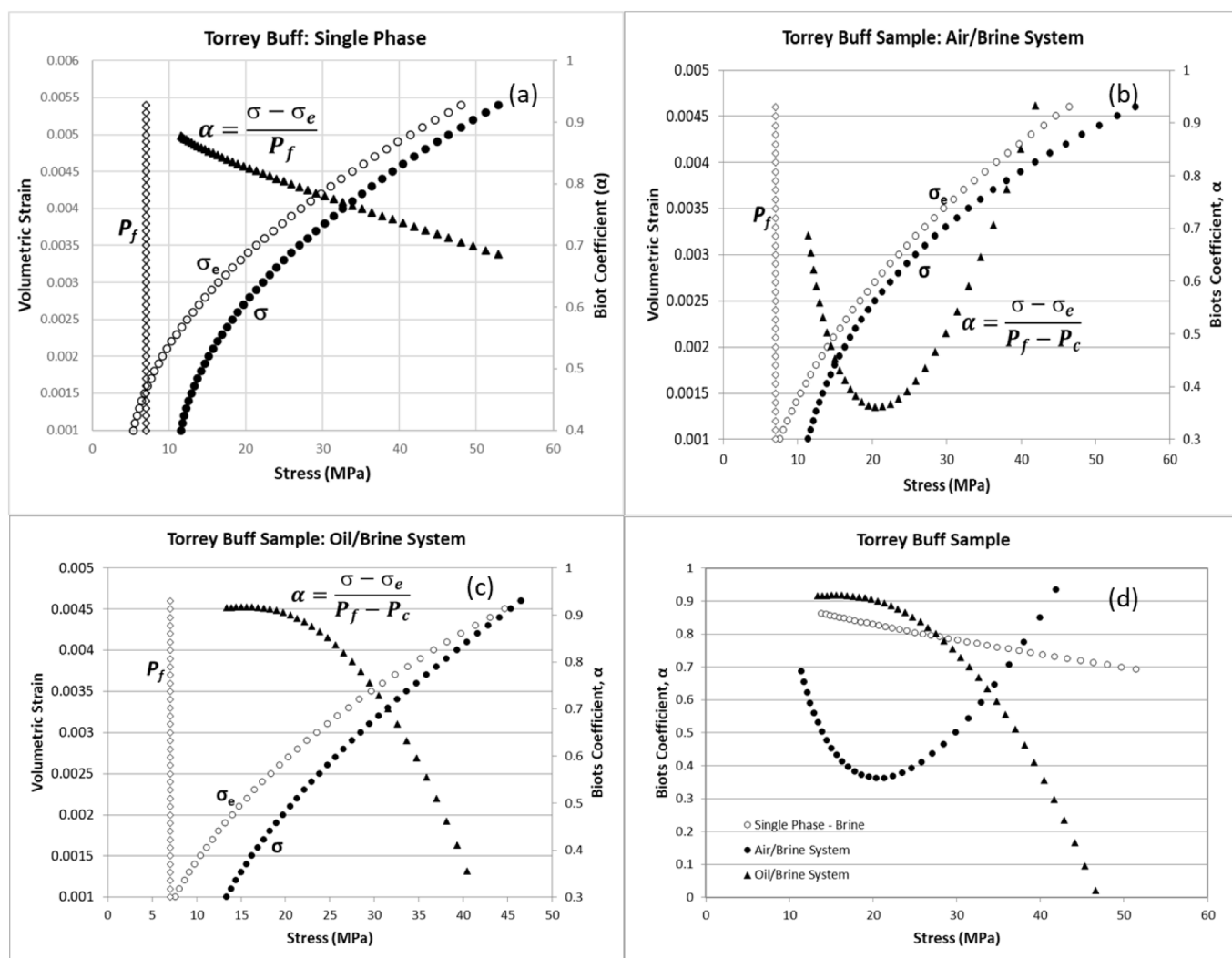


Figure 6—Measurement of effective stress, total stress and Biot coefficient for single phase brine, multi-phase air/brine and oil/brine system and their comparison for Torrey Buff Sample.

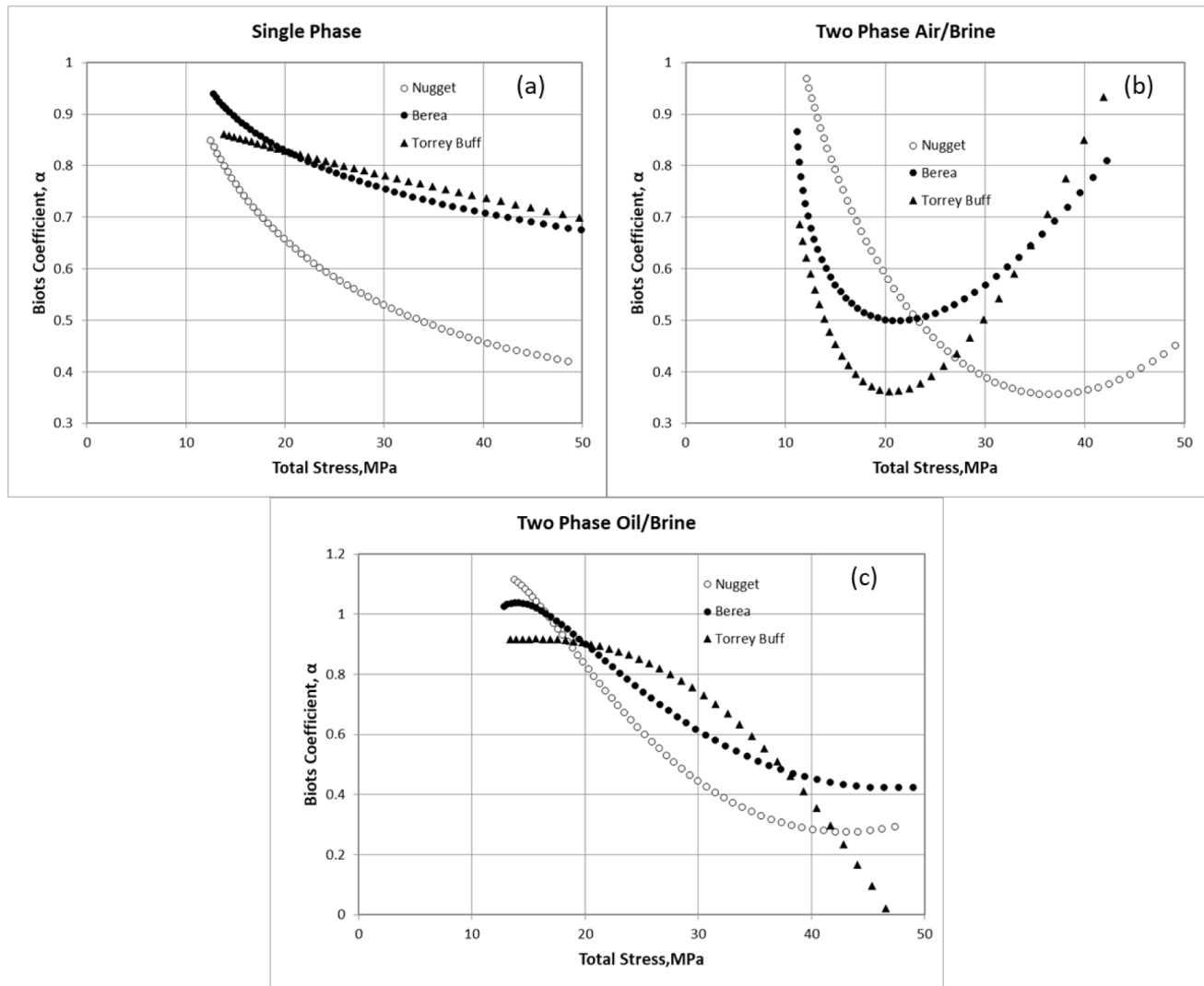


Figure 7—Comparison of Biot coefficient for the three samples during single phase and multi-phase fluid measurements

An interesting observation was the nonlinear stress dependence of Biot's coefficient in the low permeability samples of Nugget and also Berea with single phase brine, whereas the high-permeability sample (Torrey Buff) exhibited nearly linear behavior (Figure 7a). A similar behavior of linear (Blöcher et al., 2014; Ingraham et al., 2017) and nonlinear (Franquet & Abass, 1999; Zamani & Knez, 2023) Biot's coefficient vs stress was also observed in previous studies. The nonlinearity in low-permeability rocks may be attributed to the heterogeneous pore structure, where stress-induced pore closure occurs in stages. Initially, softer pores collapse at low stresses, causing a rapid initial decline in Biot coefficient. At higher stresses, only stiffer pores remain, leading to a more gradual decrease.

In contrast, high-permeability sandstones, with more uniform and stable pore networks, show a linear reduction in Biot coefficient with stress. This linearity suggests that in rocks with well-connected, larger pores, pore compressibility remains relatively constant across the tested stress range, as the dominant pore types deform uniformly under loading and the load-sharing between the solid matrix and pore fluid changes more gradually, unlike pores with high aspect ratios (Ma et al., 2024), like sheet pores (Morrow et al., 1990). It should be noted that studies on pore structure in terms of pore shapes and pore connectivity has been and remains to be one of the major challenges in fundamental petrophysics.

Non-linear Biot's Coefficient Behavior

Franquet (2020) published a vast dataset of wireline straddle packer Microfrac tests incorporating the poro-elasticity in the derivation of the in-situ stress profile. This work required the estimation (log-derived), and measurement (lab-derived), of Biot's coefficient across the intervals where the stress measurements were conducted. This dataset permits the study of Biot's coefficient against basic petrophysical properties such as: compressional and shear wave slownesses, porosity, and formation density. All cross-plot clearly indicate a strong non-linearity against these rock physics properties, as show in Figure 8, including results of Franquet & Abass (1999), Fabricius (2007), and Selvadurai & Suvorov (2020) along with Gustkiewicz empirical correlation.

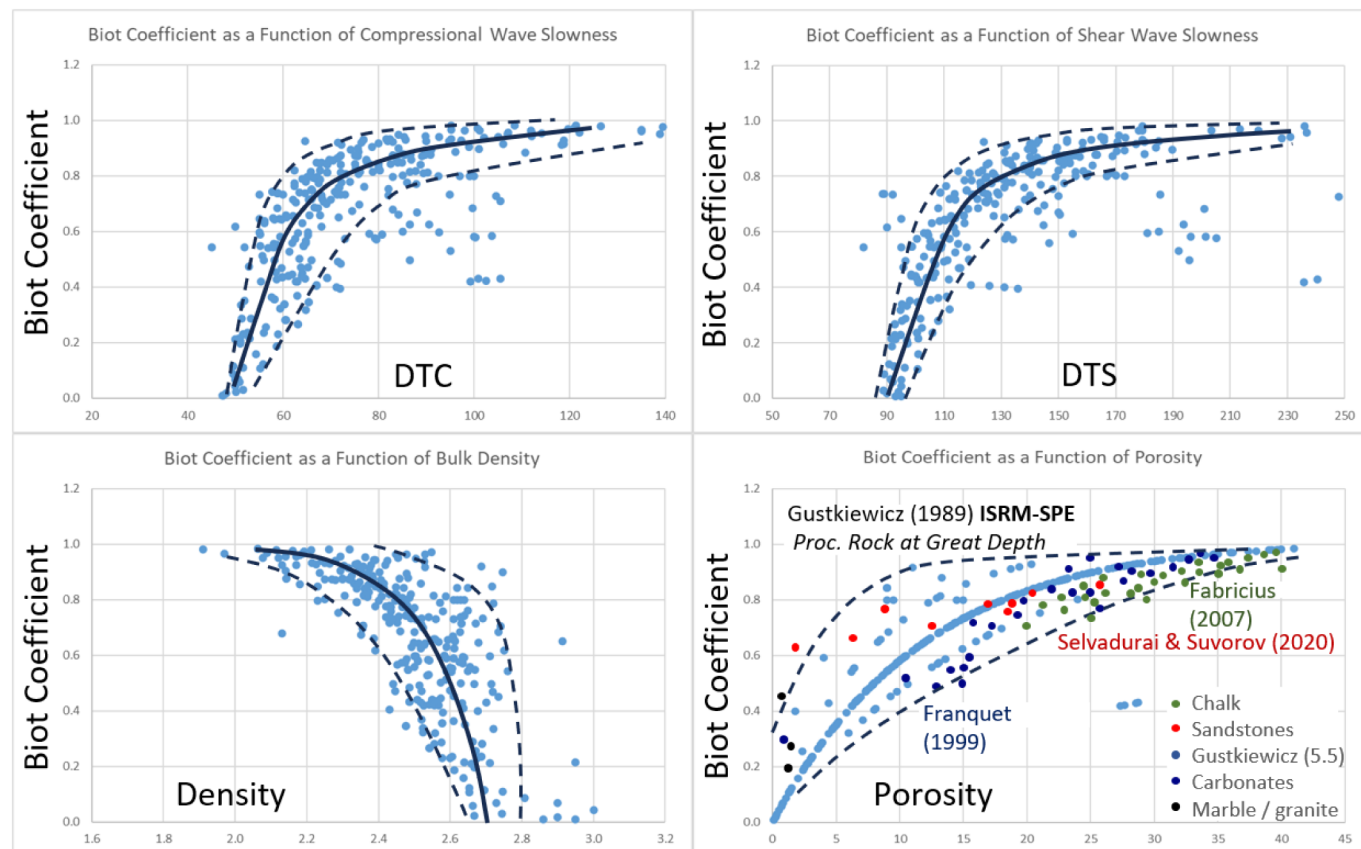


Figure 8—Non-linear behavior of Biot's coefficient against basic petrophysical rock properties.

The behavior of Biot's coefficient vs. compressional and shear acoustic slowness (DTC and DTS) shows a rapid increase from 0 to 0.8 as DTC increase from 50 to 65 $\mu\text{s}/\text{ft}$ (and DTS from 90 to 120 $\mu\text{s}/\text{ft}$) in tight formations, while it exhibit a slow increase from 0.8 to 1 as DTC continue to increase from 65 to 140 $\mu\text{s}/\text{ft}$ (and DTS from 120 to 230 $\mu\text{s}/\text{ft}$) in high porosity rocks. The Biot's coefficient behavior vs. formation density is reverse than acoustic slowness(es). Biot exhibits a slow decrease from 1 to 0.8 as the rock density increase from 2.1 to 2.5 g/cc in high porosity rocks, while it shows a rapid decrease from 0.8 to 0 as density increase from 2.5 to 2.7 g/cc . Ighodalo et al. (2025) demonstrated that the best fixing coefficient in Gustkiewicz (1989) correlation is a = 5.5 to estimate Biot's coefficient as a function of total porosity. Using the Equation 7 below where ϕ is the total porosity and a is a correlation coefficient.

$$\alpha_s = 1 - e^{-a \tan\left[\frac{\pi}{2} \phi\right]} \quad (7)$$

The stress-dependent Biot coefficient has significant implications for reservoir geomechanics. In field development, accurate prediction of effective stress is crucial for assessing compaction, subsidence, and

permeability evolution. The nonlinear behavior of Biot coefficient in low-permeability formations suggests that traditional linear poroelastic models may underestimate stress changes at early production stages, where pore collapse is most pronounced. Conversely, high-permeability reservoirs can be modeled more reliably using constant Biot coefficient values.

Effect of Capillary Pressure on Biot Coefficient

Air/Brine System. In order to quantify the effect of P_c on the Biot coefficient, experimental measurements were conducted on rock samples saturated with single-phase (brine) and two-phase fluids (air/brine and oil/brine). Figure 4b, Figure 5b & Figure 6b show effective stress and total stress measurements with increasing stress for two-phase air/brine system for the three samples respectively. Multiphase Biot coefficient is calculated from Eqn. 4 using measured values of total stress, effective stress, pore pressure and capillary pressured (about 200 psi or 1.5 MPa, Table 2) for air/brine system for all the three samples.

Also, the air/brine system exhibited complex Biot coefficient (α_m) evolution with increasing stress as shown in Figure 7b. For the tested samples, Biot coefficient (α_m) evolution with increasing stress was characterized by an initial decrease, reach a minimum value, followed by a subsequent increase.

The initial decrease in α with respect to increasing stress is expected, as the single phase Fig 7a, and is attributed to pore throat compression and reduction in porosity. Increasing confining stress leads to compression of rock pore structure. Experimental studies using techniques such as NMR, mercury intrusion porosimeter, and microstructural analysis consistently show that as confining stress increases, both pore bodies and pore throats become smaller (Al-Hamad *et al.*, 2024; Ighodalo *et al.*, 2025). This compression is especially pronounced in rocks with higher initial permeability and larger pores, which are more susceptible to stress-induced reduction in pore volume and throat size (Gao *et al.*, 2019; Shar *et al.*, 2017; Liu *et al.*, 2020). As pore throats narrow under increased confining stress, the pore fluids, in the present case air and brine could redistribute and has the possibility of altering capillary pressure, an interesting area for future studies.

As the confining stress increases, porosity decreases until a critical value is reached, beyond which, continuous increasing stress can lead to pore collapse. An increase in α after reaching a minimum value could be due to the initiation of pore collapse and pore fluid rearrangement. Beyond a critical stress, microcracks and smaller pores may begin to coalesce, and larger pores may collapse, eliminating capillary pathways. This fluid redistribution at higher stress can enhance the ability of pore fluid to share more of the applied total stress and could be the reason for the observed increase in α after reaching a minimum value. The stress at which the trend reversal is observed seem to depend on the quality of rock. For the high poro-perm Torrey Buff and Berea samples, minimum α was observed at 22 MPa confining stress, while for Nugget sample it was observed at 37 MPa.

Comparing Biot coefficients for single phase brine and the multiphase air/brine system (Figure 7), at lower total stress (confining stress), order of samples exhibiting α seem to reverse. For single phase brine measurements, Nugget sample showed least values of α (Figure 7a), while, during multiphase air/brine measurements, α for Nugget sample was the highest. For samples saturated with multiphase air/brine, as the air is compressible, it does not have the capability to share externally applied total stress and almost all the applied stress acts on the rock matrix, resulting in pore throat compression especially for Berea and Torrey Buff. Whereas Nugget is a low-quality tight rock, with a relatively stiffer rock matrix, with limited scope for pore compression, especially at low stress. Therefore, no significant changes in Biot coefficient is observed for Nugget sample as shown in Figure 4d as compared to Figure 5d & Figure 6d for Berea and Torrey Buff samples, wherein there is a significant difference in Biot coefficient between single phase (α_s) and multiphase air/brine system (α_m).

Oil/brine system. Similarly, Figure 4c, Figure 5c & Figure 6c show effective stress, total stress and Biot coefficient measurements with increasing stress for two-phase oil/brine system for the three samples

respectively. Biot coefficient is calculated from Eqn. 4 using measured values of total stress, effective stress, pore pressure and capillary pressured for air/brine system as shown in Table 2 as about 100 psi or 0.7 MPa for all the three samples. Biot coefficient (α_m) for oil/brine system mostly exhibited a decreasing trend with respect to applied stress, only starting to increase at the end of the stress cycle. In contrast to the air/brine system, Biot coefficient begin to increase only at higher stress values. Figure 7c.

A summary of the observations for the tested samples and systems at the tested conditions is described in Table 3. The behavior of Biot coefficient reaching a minimum value followed by an increase was predominantly observed for multiphase air/brine system only. No such observation was made for single phase brine system and only beginning to observe it for oil/brine system towards the end at high stress values and visible only for Nugget and Berea samples. This behavior observed mostly by air/brine system could be due to compressibility of the pore fluid. For single phase brine and oil/brine system, the compressibility of these pore fluids is almost negligible and have a better capability to share the externally applied total stress when compared to air/brine system. As air is highly compressible and as a pore fluid is unable to support the rock matrix. Most of the externally applied total stress acts on the matrix, reducing the size of pore size and pore throat. This reduction continues until a critical minimum porosity is reached beyond which pore collapse is initiated.

Table 3—Summary of observations for single phase and multiphase Biot coefficient measurements

Rock type	Nugget	Berea	Torrey Buff
Brine			
Initial Slope	-0.034	-0.02	-0.0054
$\sigma @ \alpha_{s_min}$	Not Observed	Not Observed	Not Observed
Remarks	α_s did not reach a minimum value for the applied stress for single phase brine saturated samples		
Air-brine			
Initial Slope	-0.065	-0.13	-0.069
$\sigma @ \alpha_{m_min}$	37 MPa	22 MPa	22 MPa
Remarks	Minimum value of α_m was observed at higher stress for low quality Nugget rock when compared to good quality Berea and Torrey Buff rocks.		
Oil-brine			
Initial Slope	-0.039	0.00024	-0.00047
$\sigma @ \alpha_{m_min}$	45 MPa	45 MPa	Not Observed
Remarks	Minimum value of α_m was observed at higher stress for low quality Nugget rock when compared to good quality Berea and Torrey Buff rocks.		

Conclusions

This study demonstrates the significance of capillary pressure (P_c) influence on Biot coefficient in multiphase systems. The experimental results show fundamentally different stress and Biot coefficient behaviors in air/brine versus oil/brine systems.

- Biot coefficient measurements were first performed on sandstone samples fully saturated with a single-phase fluid (brine) to establish a reference for multiphase measurements that of air/brine and oil/brine systems. Biot coefficient showed a decreasing nonlinear trend with increasing applied stress, which is an expected behavior, in confirmation with previous studies. This is attributed to the progressive closure of microcracks and pore throats and increase in grain contact under compressive load, which increases the stiffness of the rock matrix and reduction in porosity. The Biot coefficient seems to depend on the quality of rocks with the lowest value reported for Nugget sample (least permeability), compared to Berea and Torrey Buff. Also, a nonlinear stress dependence of Biot's coefficient was observed for lower permeability samples (Nugget & Berea) whereas high-permeability sample (Torrey Buff) exhibited nearly linear behavior
- Multi-phase systems, especially air/brine, exhibited complex Biot coefficient evolution with increasing stress, which was characterized by an initial decrease, reaching a minimum value, followed by a subsequent increase. The initial decrease and subsequent increase of Biot in air/brine systems is attributed to pore compression and later pore collapse.
- In contrast to air/brine system, Biot coefficient for Oil/brine system mostly exhibited a decreasing trend with respect to applied stress, only starting to increase at the end of the stress cycle.
- Generally, Biot coefficient was lower for multiphase air/brine and oil/brine systems compared to single phase brine, indicating that the capillary pressure has an effect that of compression on the rock matrix, increasing the effective stress on the rock.
- These insights have important implications for reservoir characterization, Geo-mechanical modeling in hydrocarbon recovery, and production forecasting, especially in gas reservoirs. It is also important to note the non-linearity of Biot's coefficient behavior with effective compression or porosity reduction due to triaxial loading as it is observed in Nugget and Berea sandstones plugs.
- Future work should focus on expanding the fluid system variability and incorporating in-situ imaging techniques to better understand the pore-scale processes governing these behaviors.

Nomenclature

σ	= Total Stress
σ^{UD}	= Total Stress from undrained test
σ_e^{D}	= Effective Stress from drained test
α	= Biot Coefficient
α_s	= Biot Coefficient single phase
α_m	= Biot Coefficient multiphase
DTC	= Compressional wave slowness
DTS	= Shear wave slowness
P_c	= Capillary pressure
P_f^{UD}	= Pore fluid pressure during undrained test
P_o	= Oil phase pressure
P_w	= Water phase pressure
K_{fr}	= Bulk Modulus of frame or matrix
K_s	= Bulk modulus of solid or grains
ϕ	= porosity

References

- Alam, M., & Fabricius, I., 2013. Change in Biot's effective stress coefficient of chalk during pore collapse, Poromechanics V: Proceedings of the Fifth Biot Conference on Poromechanics.
- Albannay, A., Bui, B., & Katsuki, D., 2021. The Effect of Capillary Condensation on the Geo-mechanical and Acoustic Properties of Tight Formations: An Experimental Study. Day 3 Wed, November 17, 2021.

- Al-Hamad, M., Klemin, D., Ma, S., and Abdallah, W., 2024. Insight of core analysis data interpretation by use of digital rock physics," paper SPWLA-2024-007, 2024 SPWLA 65th Annual symp., 18-22 May, Rio de Janeiro, Brazil.
- Al-Harbi, A., Schmitt, D., and Ma, S., 2011. Toward quantitative remaining oil saturation: determination challenges & techniques, paper SPE-147651, ATCE, Denver, CO, 30 Oct – 2 Nov.
- Arii H., Morita N., Ito Y., and Takano E., Waseda U., 2005. Sand-Arch Strength Under Fluid Flow With and Without Capillary Pressure, paper SPE 95812, ATCE, Dallas, Texas, U.S.A., 9 – 12 Oct.
- Biot, M. A., 1941. General theory of three-dimensional consolidation. *J. appl. Physics, Lancaster Pa.*, **12**, pp. 155–164
- Bishop AW., 1959. The principle of effective stress. *Tek Ukebl.*; **106**:859–863.
- Blöcher, G., Reinsch, T., Hassanzadegan, A., Milsch, H., & Zimmermann, G., 2014. Direct and indirect laboratory measurements of poroelastic properties of two consolidated sandstones. *International Journal of Rock Mechanics and Mining Sciences*, **67**, 191–201.
- Bui, B., & Tutuncu, A., 2015. Effect of capillary condensation on geomechanical and acoustic properties of shale formations. *Journal of Natural Gas Science and Engineering*, **26**, 1213–1221.
- Detournay, E., & Cheng, A. H.-D., 1993. Fundamentals of Poroelasticity. In *Comprehensive Rock Engineering*.
- Gao, H., Wang, C., Cao, J., He, M., & Liangbin, D., 2019. Quantitative study on the stress sensitivity of pores in tight sandstone reservoirs of Ordos basin using NMR technique. *Journal of Petroleum Science and Engineering*.
- Guerriero, V., & Mazzoli, S., 2021. Theory of Effective Stress in Soil and Rock and Implications for Fracturing Processes: A Review. *Geosciences*.
- Franquet, J. A., & Abass H. H., 1999. Experimental evaluation of Biot's poroelastic parameter- Three different methods, *Rock Mechanics for Industry*.
- Franquet, J.A., 2020. Advances of Straddle Packer Microfrac Stress Measurements for Reservoir Development – The Pursuit of Subsurface Tectonic Strain Behavior. IPTC-20257-MS, IPTC, Dhahran, Saudi Arabia. 13-15 Jan.. <https://doi.org/10.2523/IPTC-20257-MS>.
- Gustkiewicz J., 1989. Synoptic View of Mechanic Behavior of Rocks under Triaxial Compression. Proc. ISRM-SPEInternational Symposium Rock at Great Depth, Pau, France. (Edited by Maury and Fourmaintraux). Balkema, Rotterdam. pp. 3–9.
- Haghi, A., & Chalaturnyk, R., 2022. Experimental Characterization of Hydrodynamic Properties of a Deformable Rock Fracture. *Energies*.
- Hart, D. J., & Wang, H. F., 1995. Laboratory measurements of a complete set of poroelastic moduli. *JGR: Solid Earth*, **100**(B8).
- Ighodalo, E.O., Ma, S., and Van Steene, M., 2025. Pore network characterization utilizing LWD acoustic pore aspect ratio and NMR rock typing for enhanced well placement, paper SPWLA-2025-0030, SPWLA 66th ALS, 17-21 May, Dubai, UAE.
- Ighodalo, E.O., Berkane, M., Franquet, J.A., and Chakrabarti, P., 2025. Formation-Dependent Tectonic Lateral Strain Calibration for CCUS Operations Applying Microfrac Tests. International Petroleum Technology Conference, Kuala Lumpur, Malaysia. 18-20 February. IPTC-24673-MS. <https://doi.org/10.2523/IPTC-24673-MS>
- Ingraham, M., Bauer, S., Issen, K., & Dewers, T., 2017. Evolution of permeability and Biot coefficient at high mean stresses in high porosity sandstone. *International Journal of Rock Mechanics and Mining Sciences*, **96**, 1–10.
- Kasani, H., & Selvadurai, A., 2022. A Review of Techniques for Measuring the Biot Coefficient and Other Effective Stress Parameters for Fluid-Saturated Rocks. *Applied Mechanics Reviews*.
- Kümpel, H., 1991. Poroelasticity: parameters reviewed. *Geophysical Journal International*, **105**, 783–799.
- Liu, G., Yin, H., Lan, Y., Fei, S., & Yang, D., 2020. Experimental determination of dynamic pore-throat structure characteristics in a tight gas sandstone formation with consideration of effective stress. *Marine and Petroleum Geology*.
- Ma, J., Liang, L., Al-Hamad, M. et al, 2024. Aspect ratio dependent pore size distribution from MICP measurement," IPTC-23659, IPTC, 12-14 Feb, Dhahran, Saudi Arabia.
- Ma, S., Jiang, M., and Morrow, N.R., 1991. Correlation of capillary pressure relationships and calculations of permeability, SPE 22685, the 66th SPE ATCE, Dallas, Oct. 6-9.
- Ma, S., Mason, G., and Morrow, N.R., 1996. Effect of contact angle on drainage and imbibition in regular polygonal tubes, *Colloids and Surfaces, A; Physicochemical and Engineering Aspect*, v.**117**, pp.273.
- Ma, S., Al-Hajari, A., Kersey, D. et al, 2006. Enhanced petrophysics: integration of core & log data for improved reservoir saturation monitoring," SPE 106350, 2006 SPEKSA Technical Symp.
- Ma, S. and Musharfi, N., 2020. Composition and methods to recover irreducible water for enhanced formation evaluation, US Patent 10,633, **574**, 28 April.
- Mahavadi, S., Al-Hamad, M., Ma, S., and Abdallah, W., 2022. Role of Polar Species in Determining the Interfacial Tension of a Crude Oil/Water System," *Journal of Energy & Fuels*, 21 June, <https://doi.org/10.1021/acs.energyfuels.2c00557>.
- Morrow, N.R., Brower, K., Ma, S., and Buckley, J., 1990. Fluid flow in healed tectonic fractures, *JPT*, pp.1310, Oct.

- Maranha, J., Pereira, C., & Cardoso, R., 2022. Effective stress in unsaturated soils: Lessons from capillarity in regular sphere arrangements. *Geomechanics for Energy and the Environment*.
- Mavko, G., Mukerji, T., & Dvorkin, J., 2009. The Rock Physics Handbook. Cambridge University Press.
- Moghadam, A., Vaisblat, N., Harris, N., & Chalaturnyk, R., 2020. On the magnitude of capillary pressure (suction potential) in tight rocks. *Journal of Petroleum Science and Engineering*, **190**, 107133.
- Papamichos, E., Brignoli, M., & Santarelli, F., 1997. An experimental and theoretical study of a partially saturated collapsible rock. *Mechanics of Cohesive-frictional Materials*, **2**, 251–278.
- Saurabh, S., & Harpalani, S., 2018. The effective stress law for stress-sensitive transversely isotropic rocks. *International Journal of Rock Mechanics and Mining Sciences*, **101**, 69–77.
- Shar, A., Mahesar, A., Chandio, A., & Memon, K., 2017. Impact of confining stress on permeability of tight gas sands: an experimental study. *Journal of Petroleum Exploration and Production Technology*, **7**, 717–726.
- Syed, S.A., Jin, G., and Ma, S., 2022. Frequency dependent rock mechanical properties for geo-mechanical applications, paper IPTC 22309, 21-23 Feb, Riyadh, Saudi Arabia.
- Ali, S., Al-Ofi, S., and Ma, S., 2024. Relationship between rock acoustic dispersion and petrophysical properties for enhanced formation evaluation, IPTC-23747, 2024 IPTC, 12-14 Feb, Dhahran, Saudi Arabia.
- Terzaghi, K., 1936. The shearing resistance of saturated soil. *Proceeding of the First International Conference in Soil Mechanics*, **V1**:pp. 54–56.
- Terzaghi, K., 1943. Theoretical soil mechanics. J. Wiley & Sons, inc.
- Yoon, H., & Kim, J., 2023. The impacts of scaled capillary pressure combined with coupled flow and geomechanics on gas hydrate deposits. *Geomechanics for Energy and the Environment*.
- Zamani, M., & Knez, D., 2023. Experimental Investigation on the Relationship between Biot's Coefficient and Hydrostatic Stress for Enhanced Oil Recovery Projects. *Energies*.
- Zhang, Y., & Yang, D., 2024. A two-phase flow model coupled with geomechanics for simulating transient production behaviour in an unconventional reservoir by considering fracture geometry and capillary pressure. *Fuel*.
- Zimmerman, R.W., 1991. Compressibility of Sandstones, 1st Edition, Volume **29**.